Hence, the relation may be given as

$$y = kx \quad \Rightarrow \quad y = \frac{5}{6}x$$

(ii) From the graph, $y = \frac{5x}{6}$, if $x = 90$, then $y = \frac{5}{6} \times 90 = 75$ km

The distance travelled for $1\frac{1}{2}$ hours (i.e.,) 90 minutes is 75 km.

(iii) From the graph, $y = \frac{5x}{6}$, if $y = 300$ then $x = \frac{6y}{5} = \frac{6}{5} \times 300 = 360$ minutes (or) 6 hours.

The time taken to cover 300 km is 360 minutes, that is 6 hours.

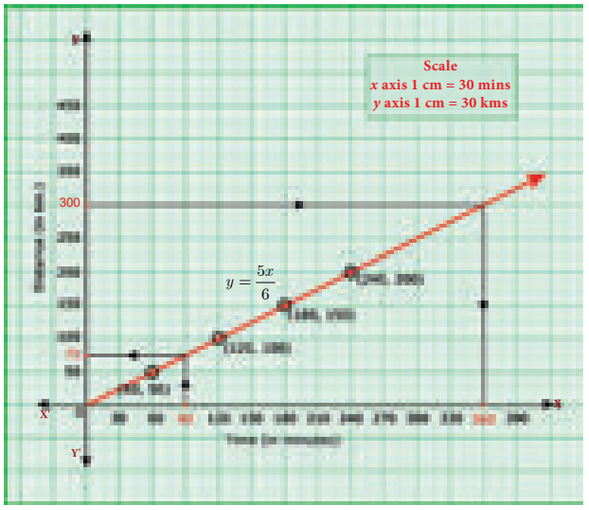


Fig. 3.11

(ii) Indirect variation:

The distance between Chennai and Madurai is (nearly) 480 km. Think of a train that starts from Chennai and travels towards Madurai. As it **increases** speed more and more, the time taken for travel will **decrease**. In the following table speed v is given in km and time t is given in hours:

Speed (v) (km/hr)	30	40	60	80
Time (t) (hours)	16	12	8	6

From the table it is clear that if you travel at a slower speed, the time increases and if the train is faster, the time decreases. You find, 30×16 = 40×12 = 60×8 = 80×6, which tells that vt is a constant. Here $vt = 480$. In such a case, we say the variables v and t are inversely proportional. Observe that the graph of equation like $vt = 480$ will not be a straight line. Inverse variation implies that as one variable increases, the other variable decreases.

Visualising Indirect variation:

Look at the adjacent graph. It is a graph of the equation $xy = 8$. We have taken only the [positive values of x, y.

The table of values is:

x	1	2	4	8
$y = \frac{8}{x}$	8	4	2	1

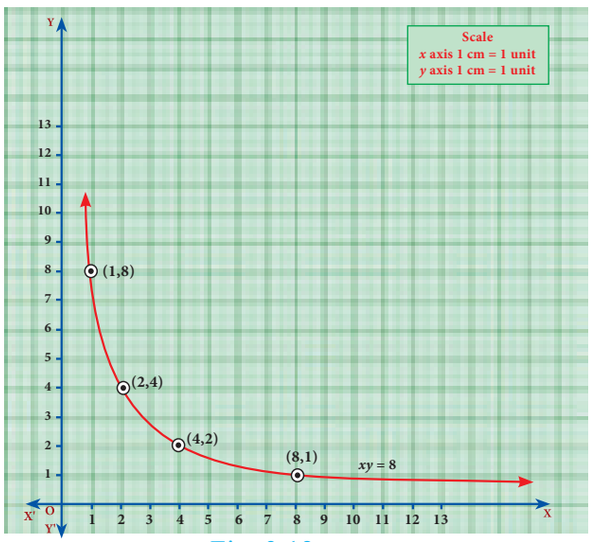


Fig. 3.12

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